Harshil Chauhan

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GitHub | LinkedIn | Portfolio

Professional Summary

Mechanical Engineering undergraduate with hands-on research experience in robotics, autonomous underwater vehicle (AUV) simulation, and control systems. Skilled in ROS 2, Gazebo, ArduSub, ArduPilot, MATLAB/Simulink, Python, and SQL, with project experience spanning computer vision and cloud data engineering (AWS, Snowflake, DBT). Proven leadership as Training & Placement Coordinator for 180+ students.

Education

National Institute of Technology, Durgapur Aug 2024 – May 2028 (Expected)
Bachelor of Technology (B.Tech) in Mechanical Engineering CGPA: 8.22/10
Class XII (CBSE, 2024): 90.08% · **Class X** (CBSE, 2022): 94.80%

Technical Skills

Programming Languages: Python, C++, C, MATLAB, SQL
Robotics & Simulation: ROS 2, Gazebo Harmonic, ArduSub, ArduPilot, Simulink, SolidWorks, Blender
Control Systems & Perception: Cascaded PID Control, Trajectory Tracking, Depth Control, Computer Vision, OpenCV
Tools & Platforms: Git, GitHub, Linux, AWS S3, Snowflake, DBT Core/CLI, Visual Studio Code, Cursor
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, System Design, Data Modeling (OBT, Star Schema)
Soft Skills: Leadership, Problem Solving, Team Collaboration, Communication, Project Management

Experience

Research Intern – Marine Robotics & Autonomous Systems May 2026 – July 2026
CAIR, Indian Institute of Technology (IIT) Mandi Advisor: Prof. Jagadeesh Kadiyam

- Researched AUV simulation using **ROS 2, Gazebo Harmonic, and ArduSub** for underwater inspection and navigation.
- Integrated perception, localization, and control modules with simulated **camera, DVL, and sonar** sensors.
- Developed and tested ROS 2 nodes for sensor communication, image processing, and autonomous vehicle control.
- Designed **cascaded PID control** for depth and trajectory tracking; exploring adaptive and learning-based tuning.
- Collaborated with research mentors to validate simulation results and improve navigation algorithm robustness.

Projects

Autonomous Underwater Vehicle Navigation System ROS 2, Gazebo Harmonic, ArduPilot, OpenCV, Python

- Built an AUV simulation framework using **ROS 2, Gazebo Harmonic, and ArduPilot** to evaluate autonomous navigation.
- Integrated **OpenCV** path detection and guidance commands for autonomous waypoint following.
- Designed and validated cascaded PID depth and trajectory control under varying simulated conditions.
- Implemented ROS 2 nodes for sensor data processing, vehicle control, and inter-node communication.

NexusData Warehouse AWS S3, Snowflake, DBT, Jinja

- Architected an end-to-end data engineering pipeline implementing medallion architecture (Bronze, Silver, Gold layers) for processing large-scale transactional and operational data.
- Configured secure data ingestion from **AWS S3** data lakes to **Snowflake** utilizing external stages, file formats, and IAM security protocols.
- Utilized **DBT Core (CLI)** and **Jinja templates/macros** to build dynamic, parameter-driven transformation pipelines and implement incremental data loading strategies.
- Designed downstream analytical data models including One Big Table (OBT) and Star Schema architectures with Slowly Changing Dimensions (SCD Type 2).
- GitHub: github.com/harshil28006

AI-Based UAV Threat Detection System (Tata Technologies InnoVent Hackathon) MATLAB, Simulink

- Developed a MATLAB/Simulink flight model with waypoint navigation, trajectory tracking, and real-time control.
- Implemented **cascaded PID** flight control for stable attitude and position regulation in simulation.
- Designed the architecture for future ROS 2, Gazebo, and edge AI integration for aerial threat detection.

Leadership Experience

Training & Placement Coordinator – Department of Mechanical Engineering, NIT Durgapur Mar 2026 – Present

- Primary liaison between the Training & Placement Cell, faculty, students, and recruiters.
- Coordinate drives, eligibility, and interview schedules for **180+ students**; assist recruiters with logistics.

Senior Coordinator – SPIC MACAY, NIT Durgapur Aug 2025 – Present

- Organized cultural heritage events, including the flagship **Virasat** program with **200+** student participants.
- Managed logistics, volunteers, and stage; supported institute events including **Aarohan** and **Recstacy**.